



COMPETENCY STANDARD

FOR

SHIP SAFETY AND FIREFIGHTING

(SHIPBUILDING SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for Ship Safety and Firefighting serve as a foundational document for the development of curricula, teaching materials, and assessment tools specifically tailored to the shipbuilding sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant roles within the sector.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP), this document addresses the skills requirements critical to Ship Safety and Firefighting. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance,
Probashi Kallyan Bhaban (Level-15 & 16)
71-72 Eskaton Garden, Dhaka-1000
Phone: +8802 55138753-55,
Website: <https://sicip.gov.bd/>

INTRODUCTION:

The **Skills for Industry Competitiveness and Innovation Program (SICIP)** has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from Construction industry in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:**Generic Competencies (40 Hrs.)**

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SBD-SSF-01-G	Apply Occupational Health and Safety (OHS) Practice in the Workplace	1. Identify OHS policies and procedures. 2. Apply personal health and safety practices. 3. Report hazards and risks 4. Respond to emergencies.	20
SICIP-SBD-SSF-02-G	Operate in a Team Environment	1 Identify team goals and work processes. 2 Identify own role and responsibilities within team. 3 Communicate and co-operate with team members. 4 Practice problem solving within the team.	20
Total Hour			40 Hrs.

Sector Specific Competencies (Total 40 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SBD-SSF-01-S	Apply Green Practices in Ship Building Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	20
SICIP-SBD-SSF-02-S	Work with Hand Tools and Power Tools	1. Identify and inspect hand and power tools 2. Use hand tools properly and safely 3. Operate power tools properly and safely. 4. Clean and maintain hand and power tools	20
Total Hour			40 Hrs.

Occupation Specific Competencies (Total 280 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-SBD-SSF-01-O	Understand Ship Structure and SOLAS Convention	1. Identify major structural parts of a ship 2. Identify SOLAS Convention 3. Recognize key parts of fire safety systems (FSS) code	20
SICIP-SBD-SSF-02-O	Follow Safety Procedures in Hazardous Operations	1. Use personal protective equipment (PPE) 2. Identify fire, chemical, environmental and electrical hazards 3. Apply safety rules for fire and chemicals 4. Check electrical wiring and report faults	20
SICIP-SBD-SSF-03-O	Operate Life Saving Appliances (LSA)	1. Identify types of life saving appliances 2. Demonstrate life jackets and immersion suits 3. Demonstrate emergency signaling devices 4. Explain launching procedures of lifeboats and life rafts	60
SICIP-SBD-SSF-04-O	Operate Fire Fighting Appliances (FFA)	1. Identify common causes of shipboard fires 2. Select appropriate appliances based on fire type 3. Operate portable and fixed fire extinguishers 4. Use hoses, nozzles and hydrants	60
SICIP-SBD-SSF-05-O	Operate Fire Detection and Alarm Systems	1. Identify fire detection systems 2. Test alarm panels and devices 3. Reset detection systems after testing	60

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-SBD-SSF-06-O	Respond to Fire Emergencies	1. Detect and respond to early signs of fire 2. Follow emergency fire response procedures 3. Rescue individuals from fire-affected areas 4. Report and document fire incidents	60
Total Hours			280 Hrs.

COMPETENCY STANDARD: SHIP SAFETY AND FIREFIGHTING

Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-SSF-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	2.1 OHS policies and procedures are interpreted in the workplace including <u>personal protective equipment (PPE)</u> . 2.2 Common health issues are recognized. 2.3 Common safety issues are identified and applied.
3. Report hazards and risks	3.1 Hazards and risks are identified. 3.2 Hazards and risks assessment and controls are interpreted 3.3 Hazards and risk assessment are reported
4. Respond to emergencies	4.1 Respond to alarms and warning devices. 4.2 <u>Emergency response plans and procedures</u> are responded to. 4.3 <u>First aid procedures</u> during emergency situations are identified.

Range of variables:

Variables	Range (may include but not limited to)
1. OHS policies	1.1 Organizational OHS policies 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Personal protective equipment	2.1 Safety glasses 2.2 Ear plugs 2.3 Gloves

	2.4 Apron 2.5 Helmet 2.6 Mask 2.7 Safety shoes
3. Emergency response plans and procedures	3.1 Firefighting procedures 3.2 Earthquake response procedures 3.3 Emergency response plans and procedures 3.4 Medical and first aid
4. First aid procedure	4.1 Washing of open wound 4.2 Washing chemically infected area 4.3 Applying bandage 4.4 Taking appropriate medicine

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	2.1 Identifying OHS policies and procedures 2.2 Applying personal health and safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergencies
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace documents, signs and symbols 4.3 Codes of conduct 4.4 Projector 4.5 Learning manual 4.6 Tools, equipment and facilities appropriate to the

	process or activities. 4.7 Materials are relevant to the proposed activity.
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified OHS policies and procedures 1.2 applied personal health and safety practices (including PPE) 1.3 reported hazards and risks 1.4 responded to emergencies.
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE IN A TEAM ENVIRONMENT	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-SSF-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate in a team environment. It specifically includes the tasks of identifying team goals and work processes, identifying own role and responsibilities within team, communicating and cooperating with team member, and practicing problem solving within the team.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1. Roles and objectives of the team are identified and interpreted. 1.2. Roles and responsibilities of team members and work processes are identified and interpreted.
2. Identify own role and responsibilities within team	2.1. Personal role and responsibilities are identified within the team environment. 2.2. Importance of relationships within and outside the team are interpreted .
3. Communicate and co-operate with team members	3.1. Other teammates' tasks are identified and support provided when requested. 3.2. The team is encouraged through <u>sharing information</u> or expertise, working together to solve problems, and putting team success first. 3.3. Views and opinions of other team members are interpreted and respected.
4. Practice problem solving within the team	4.1. Problems faced at the individual and team level are identified and showed insight into the root-causes of the problems. 4.2. A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3. The good ideas of others to help develop solutions are recognized and advice sought from those who have solved similar problems. 4.4. It is looked beyond the obvious and not stopped at the first answers.

Range of variables:

Variables	Range (may include but not limited to)
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1. Sharing information	1.1. Agenda 1.2. Minutes 1.3. Progress and incident reports 1.4. Operational manuals 1.5. Visual and graphic materials 1.6. Emails and SMS 1.7. Policy, procedure and standards 1.8. OHS information
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Curricular Content Guide

1. Underpinning knowledge	1.1. Team goals and work processes 1.2. Roles and responsibilities of the team 1.3. Finding problems and solving them 1.4. Importance of views and opinions of other team members
2. Underpinning skill	2.1. Identifying own role and responsibilities within team 2.2. Communicating and co-operating with team members 2.3. Demonstrating problem solving within the team
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Eagerness to learn 3.5. Tidiness and timeliness 3.6. Environmental concerns 3.7. Respect for rights of peers and seniors at workplace 3.8. Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 4.1. Workplace (simulated or actual) 4.2. Workplace documents, Projector 4.3. Learning manual 4.4. Tools, equipment and facilities appropriate to the process or activities. 4.5. Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified own role and responsibilities within team 1.2 Communicated and co-operated with team members 1.3 Demonstrated problem solving within the team
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)

3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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The Sector Specific Competencies

Unit of Competency: APPLY GREEN PRACTICES IN SHIPBUILDING SECTOR	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-SSF-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in shipbuilding sector. It specifically includes the tasks of interpreting green concepts, minimizing resources in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Green concepts in the shipbuilding sector</u> are understood. 1.2 <u>Energy-efficient technologies</u> are recognized. 1.3 <u>Use of eco-friendly materials</u> is understood. 1.4 <u>Pollution control measures</u> are interpreted. 1.5 <u>Regulatory compliance</u> with regulations and conventions is identified. 1.6 <u>Energy recovery systems</u> are identified as methods to improve energy efficiency.
2. Minimize resources use in the workplace	2.1 <u>Resource usage in shipbuilding processes</u> is identified. 2.2 Water consumption methods are identified through the installation of efficient water management systems. 2.3 <u>Process improvements</u> are understood. 2.4 Compliance with environmental regulations and industry standards is interpreted.
3. Implement waste management practices	3.1 <u>The types of waste in shipbuilding</u> are identified and described. 3.2 <u>Environmental impacts of improper waste disposal</u> in shipbuilding are understood. 3.3 <u>Principles of waste hierarchy</u> are understood for sustainable waste management. 3.4 <u>Importance of safe storage and containment of waste materials</u> is understood.

Range of Variables

Variable	Range May include but not limited to:
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1. Green concepts in the shipbuilding sector	1.1 Eco-friendly Hull Design 1.2 Use of Sustainable and Recyclable Materials 1.3 Waste Reduction and Management Practices 1.4 Emission Control and Air Pollution Reduction 1.5 Ballast Water Treatment and Marine Ecosystem Protection
2. Energy-efficient technologies	2.1 Hybrid and Electric Propulsion Systems 2.2 Advanced Hull Coatings to Reduce Drag 2.3 Waste Heat Recovery Systems 2.4 Air Lubrication Systems 2.5 Energy-efficient LED Lighting and Automation 2.6 Smart Energy Management and Monitoring Systems
3. Use of eco-friendly materials	3.1 Recycled Steel and Aluminum 3.2 Biodegradable Composites and Polymers 3.3 Sustainably Sourced Timber 3.4 Non-toxic Anti-fouling Coatings 3.5 Natural Fiber Reinforced Plastics 3.6 Low VOC (Volatile Organic Compounds) Paints and Sealants
4. Pollution control measures	4.1 Prevention of water 4.2 Prevention of air 4.3 Prevention of noise pollution
5. Regulatory compliance	5.1 International Maritime Organization (IMO) Regulations 5.2 Environmental Protection Agency (EPA) Standards 5.3 Marine Pollution (MARPOL) Convention Compliance 5.4 Occupational Health and Safety (OHS) Regulations 5.5 Ship Recycling and Basel Convention Guidelines 5.6 Local and National Environmental Permitting
6. Energy recovery systems	6.1 Waste Heat Recovery Boilers (WHRB) 6.2 Thermoelectric Energy Conversion Systems 6.3 Exhaust Gas Recirculation (EGR) Systems 6.4 Regenerative Braking for Hybrid Propulsion
7. Resource usage in shipbuilding processes	7.1 Raw Material Consumption (Steel, Aluminum, Composites) 7.2 Water Usage in Fabrication and Painting 7.3 Energy Consumption During Manufacturing 7.4 Use of Recycled and Sustainable Materials
8. Process improvements	8.1 Automation and Robotics in Assembly 8.2 Modular Construction and Prefabrication 8.3 Digital Twin and 3D Modeling Integration 8.4 Just-In-Time (JIT) Inventory Management 8.5 Enhanced Quality Control and Testing Procedures
9. Types of waste in shipbuilding	9.1 Hazardous wastes 9.1.1 Paints 9.1.2 Solvents 9.1.3 Oils 9.1.4 Asbestos

	9.2 Non-hazardous wastes 9.2.1 Metal scraps 9.2.2 Wood and general refuse
10. Environmental impacts of improper waste disposal	10.1 Including soil contamination 10.2 Water pollution, and air quality degradation, are understood.
11. Principles of waste hierarchy	11.1 Reduction 11.2 Reuse 11.3 Recycling 11.4 Recovery 11.5 Disposal
12. Importance of safe storage and containment of waste materials	12.1 Prevent leaks 12.2 Prevent spills 12.3 Environmental contamination

Curricular Content Guide

1. Underpinning Knowledge	1.1 Fundamental Green Concepts and Sustainability Principles in Shipbuilding 1.2 Overview of Energy-Efficient Technologies and Their Applications 1.3 Types and Benefits of Eco-Friendly Materials in Shipbuilding 1.4 Key Pollution Control Measures and Their Implementation 1.5 Regulatory Frameworks, Conventions, and Compliance Requirements 1.6 Energy Recovery Systems and Their Role in Enhancing Efficiency 1.7 Resource and Water Usage Management in Shipbuilding Processes 1.8 Understand of environmental Impacts, and Sustainable Practices
2. Underpinning Skills	2.1 Applying Green Concepts in Shipbuilding Practices 2.2 Evaluating and Selecting Energy-Efficient Technologies 2.3 Identifying and Using Eco-Friendly Materials Effectively 2.4 Implementing Pollution Control Measures Onsite 2.5 Interpreting and Ensuring Regulatory Compliance 2.6 Operating and Maintaining Energy Recovery Systems 2.7 Monitoring and Managing Resource and Water Usage 2.8 Improving Shipbuilding Processes for Environmental Sustainability 2.9 Classifying, Handling, and Safely Storing Shipbuilding Waste

3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimizing resources use in the workplace 1.3 implementing waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: USE HAND AND POWER TOOLS	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-SSF-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to use hand and power tools in the workplace. It specifically includes the task of identifying and inspecting hand and power tools, using hand tools properly and safely, operating power tools properly and safely and cleaning and maintaining hand and power tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and inspect hand and power tools	1.1 Appropriate hand and power tools are identified. 1.2 Application of hand and power tools is recognized. 1.3 Usability of hand and power tools is checked and verified.
2. Use hand tools properly and safely	2.1 Appropriate <u>hand tools</u> are selected. 2.2 Safety precautions are ensured before using hand tools. 2.3 Unsafe or faulty hand tools are identified and marked for repair. 2.4 <u>Measuring tools</u> are checked and calibrated before use. 2.5 Use hand tools properly and safely to perform work activity.
3. Operate power tools properly and safely	3.1 Appropriate <u>power tools</u> are selected. 3.2 Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.3 Safety precautions are ensured before using power tools in accordance with manufacturer's operating specification. 3.4 Proper sequence of operation applied for using power tools. 3.5 Unsafe or faulty power tools are identified and marked for repair. 3.6 Operate power tools properly and safely to perform work activity.
4. Clean and maintain hand and power tools	4.1. Dust and foreign matter are removed from hand and power tools in accordance to workplace standards. 4.2. Condition of hand and power tools is checked after use and reported. 4.3. Appropriate lubricant is applied after use and prior to storage.

	4.4. Defective hand and power tools are inspected and repaired or replaced. 4.5. Hand and power tools are stored and secured in accordance with workplace requirements.
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Range of Variables

Variable	Range May include but not limited to:
1. Hand tools	1.1 Hammer 1.2 Bench vice 1.3 Files 1.4 Wrenches 1.5 Pliers 1.6 Scriber 1.7 Scraper Screw 1.8 Surface plate 1.9 Gauge 1.10 Tap sets 1.11 Die sets 1.12 Hacksaw 1.13 Spanners 1.14 Vice grip 1.15 Wire cutters 1.16 Clamps 1.17 Jacks
2. Power tools	2.1 Drills 2.2 Rivet gun 2.3 Grinders 2.4 Pneumatic wrenches 2.5 Press machine 2.6 Cutting 2.7 Saws 2.8 Soldering iron
3. Measuring tools	3.1 Micrometer 3.2 Measuring tape 3.3 Hose level 3.4 Water level 3.5 Calipers 3.6 Steel rule 3.7 Meter rule 3.8 Spirit level 3.9 Protractor 3.10 Tri-square 3.11 Gauges

Curricular Content Guide

1. Underpinning Knowledge	1.1 Information on hand and power tools, their functions and use 1.2 Procedures for safely using hand and power tools
2. Underpinning Skills	2.1 Identifying hand, power and measuring tools 2.2 Following safety precautions when using hand, power and measuring tools 2.3 Using hand and measuring tools correctly and safely in accordance with manufacturer's operating specification 2.4 Operating power tools correctly and safely in accordance with manufacturer's operating specification 2.5 Cleaning and maintaining hand and power tools after use 2.6 Applying appropriate lubricant on hand and power tools after use and prior to storing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected appropriate hand and power tools for work to be performed 1.2 identified and used measuring and testing tools appropriate to work activity 1.3 followed safety precautions when using hand and power tools
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	1.4 operated power tools safely and pursuant to manufacturer's operating specification 1.5 performed cleaning and maintenance of hand and power tools after use and prior to storing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: UNDERSTAND SHIP STRUCTURE AND SOLAS CONVENTION	Nominal Duration: 20 Hrs.	Unit Code: SICIP-SBD-SSF-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to understand ship structure and SOLAS convention. It specifically includes the task of identifying major structural parts of a ship, identifying SOLAS convention and recognizing key parts of fire safety systems (FSS) code.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify major structural parts of a ship	1.1. <u>Major structural parts of the ship</u> are identified from ship layout diagrams or physical observation. 1.2. Functions of key structural components are described. 1.3. Access routes are identified for their role in emergency evacuation and fire response.
2. Identify SOLAS Convention	2.1 <u>SOLAS Convention</u> is identified as the international standard for ship safety. 2.2 Main objective of SOLAS is explained. 2.3 SOLAS rules on fire and safety are interpreted.
3. Recognize key parts of fire safety systems (FSS) code	3.1 Fire detection and alarm systems are identified as per FSS Code requirements. 3.2 <u>Portable fire extinguishers</u> are recognized. 3.3 <u>Fixed fire extinguishing systems</u> are identified as per ship area and equipment protection. 3.4 Fire hydrants, hoses, and nozzles are recognized as part of the ship's fire main system. 3.5 Emergency escape breathing devices (EEBD) and firefighter's outfits are recognized.

Range of Variables

Variable	Range (Includes but not limited to):
1. Major structural parts of the ship	1.1 Hull 1.2 Keel 1.3 Main deck 1.4 Accommodation 1.5 Gangway 1.6 Hatch and hatch cover 1.7 Forecastle deck 1.8 Anchor winch 1.9 Anchor

	1.10 Bulkheads 1.11 Superstructure 1.12 Deck crane 1.13 Mast 1.14 Bridge 1.15 Funnel 1.16 Rudder 1.17 Propeller 1.18 Shaft
2. SOLAS convention	2.1 Chapter II-2 – Fire Protection, Fire Detection, and Fire Extinction 2.2 Chapter III – Life-Saving Appliances and Arrangements 2.3 Chapter IX – Management for the Safe Operation of Ships
3. Portable fire extinguishers	3.1 Water (APW - Air Pressurized Water) Fire Extinguisher 3.2 Foam (AFFF - Aqueous Film Forming Foam) Fire Extinguisher 3.3 Carbon Dioxide (CO ₂) Fire Extinguisher 3.4 Dry Chemical Powder (DCP or ABC Powder) Fire Extinguisher
4. Fixed fire extinguishing systems	4.1 Fire main 4.2 Sprinklers 4.3 Foam 4.4 CO ₂ 4.5 Hyper mist

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify major structural parts of the ship from ship layout diagrams or physical observation. 1.2 Describe functions of key structural components. 1.3 Identify access routes and explain their role in emergency evacuation and fire response procedures. 1.4 Identify SOLAS Convention. 1.5 Explain the main objectives of SOLAS. 1.6 Interpret SOLAS rules on fire and safety. 1.7 Identify fire detection and alarm systems. 1.8 Recognize portable fire extinguishers. 1.9 Identify fixed fire extinguishing systems. 1.10 Recognize fire hydrants, hoses, and nozzles.
2. Underpinning Skills	2.1 Performing identification of major structural parts of a ship from layout diagrams or physical observation. 2.2 Demonstrating description of the functions of key structural components. 2.3 Applying knowledge to identify access routes for emergency evacuation and fire response. 2.4 Recognizing SOLAS Convention as the international standard for ship safety.

	2.5 Performing identification of fire detection and alarm systems as per FSS Code requirements. 2.6 Recognizing portable fire extinguishers on board. 2.7 Identifying fixed fire extinguishing systems according to ship area and equipment protection. 2.8 Demonstrating recognition of emergency escape breathing devices (EEBD) and firefighter's outfits.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety. 3.2 Promptness in carrying out activities. 3.3 Sincere & honest to duties. 3.4 Eagerness to learn. 3.5 Tidiness & timeliness. 3.6 Environmental concerns. 3.7 Respect for rights of peers and seniors at workplace. 3.8 Communication with peers and seniors at workplace
4 Resource Implications	The following resources must be provided: 4.1 workplace (actual or simulate) 4.2 materials relevant to proposed activities 4.3 major structural parts of the ship 4.4 SOLAS convention

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified major structural parts of a ship 1.2 identified SOLAS Convention 1.3 recognized key parts of fire safety systems (FSS) code
2. Methods of Assessment	2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: FOLLOW SAFETY PROCEDURES IN HAZARDOUS OPERATIONS	Nominal Duration: 20 Hrs.	Unit Code: SICIP-SBD-SSF-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to follow safety procedures in hazardous operations. It specifically includes the task of using personal protective equipment (PPE), identifying fire, chemical, environmental and electrical hazards, applying safety rules for fire and chemicals and checking electrical wiring and reporting faults.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Use personal protective equipment (PPE)	1.1 Appropriate <u>Personal Protective Equipment (PPE)</u> is identified and selected. 1.2 PPE is used and worn consistently throughout the task to ensure personal safety. 1.3 PPE compliance is maintained as per organizational safety policies and occupational health regulations.
2. Identify fire, chemical, environmental and electrical hazards	3.1 <u>Fire hazards</u> are identified in accordance with onboard safety procedures. 3.2 <u>Chemical hazards</u> are recognized based on labels, MSDS, and storage conditions. 3.3 <u>Environmental hazards</u> are identified in compliance with environmental safety protocols. 3.4 <u>Electrical hazards</u> are recognized following standard electrical safety practices. 3.5 Warning signs and hazard symbols are interpreted to aid in quick identification of risks.
4. Apply safety rules for fire and chemicals	3.1 Fire safety rules are followed as per shipboard safety procedures. 3.2 Chemical safety guidelines are identified according to Material Safety Data Sheets (MSDS). 3.3 Flammable and reactive materials are interpreted and handled in designated safe areas. 3.4 Spill control measures and emergency shutdown procedures are practiced.
5. Check electrical wiring and report faults	5.1 Electrical wiring is visually inspected for signs of damage, wear, or overheating. 5.2 Loose connections, exposed wires, or burnt components are identified and recorded. 5.3 Faulty wiring or components are tagged or isolated as per safety procedures. 5.4 Reports are submitted to the supervisor or relevant authority in accordance with workplace protocols.

Range of Variables

Variable	Range (Includes but not limited to):
1. Fire hazards	1.1 Flammable liquids 1.2 Faulty or exposed wiring 1.3 Improper storage of combustible materials 1.4 Hot work (welding, cutting, grinding) without proper permits 1.5 Unattended cooking appliances 1.6 Smoking in non-designated areas 1.7 Malfunctioning electrical equipment 1.8 Improper use or storage of gas cylinders 1.9 Poor housekeeping (accumulation of waste, oily rags, etc.)
2. Chemical hazards	2.1 Toxic substances 2.2 Corrosive chemicals 2.3 Flammable chemicals 2.4 Reactive chemicals 2.5 Chemical waste
3. Environmental hazards	3.1 Oil spills 3.2 Chemical leaks or discharges 3.3 Ballast water contamination 3.4 Air pollution (e.g., exhaust emissions, SOx/NOx) 3.5 Waste mismanagement (solid or liquid) 3.6 Sewage discharge into marine waters 3.7 Plastic and microplastic pollution 3.8 Hazardous material disposal
4. Electrical hazards	4.1 Exposed live wires 4.2 Overloaded circuits 4.3 Faulty or damaged electrical equipment 4.4 Improper grounding 4.5 Wet conditions near electrical installations 4.6 Lack of proper insulation 4.7 Arcing and short circuits 4.8 Improper use of extension cords 4.9 Electrical equipment in flammable or explosive atmospheres 4.10 Loose connections 4.11 Inadequate maintenance of electrical systems
5. Personal Protective Equipment (PPE)	5.1 Apron 5.2 Helmet 5.3 Mask 5.4 Safety goggles 5.5 Ear plug 5.6 Safety hand gloves 5.7 Safety shoes

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify fire hazards in compliance with onboard safety procedures and protocols to mitigate fire risks. 1.2 Recognize chemical hazards by interpreting labels, Material Safety Data Sheets (MSDS). 1.3 Identify environmental hazards in accordance with environmental safety protocols. 1.4 Recognize electrical hazards by adhering to standard electrical safety practices. 1.5 Interpret warning signs and hazard symbols for quick and accurate identification of potential risks. 1.6 Identify chemical safety guidelines and procedures based on the Material Safety Data Sheets (MSDS) 1.7 Identify loose connections, exposed wires, or burnt components, marking them for attention.
2. Underpinning Skills	2.1 Demonstrating the correct use and consistent wearing of PPE throughout tasks. 2.2 Identifying fire hazards in accordance with onboard safety procedures. 2.3 Recognizing chemical hazards using labels, MSDS, and proper storage conditions. 2.4 Recognizing electrical hazards and interpreting warning signs and hazard symbols. 2.5 Applying fire safety rules and chemical safety guidelines according to shipboard procedures. 2.6 Performing spill control measures and emergency shutdown procedures. 2.7 Inspecting electrical wiring for damage, wear, or overheating. 2.8 Fixing loose connections, exposed wires, or burnt components by tagging or isolating them. 2.9 Maintaining reports of faulty wiring or components and submitting them to the supervisor.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal Protective Equipment (PPE) 4.3 Chemical safety guidelines

	4.4 Material and Safety Data Sheet
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 used personal protective equipment (PPE) 1.2 identified fire, chemical, environmental and electrical hazards 1.3 applied safety rules for fire and chemicals 1.4 checked electrical wiring and report faults
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE LIFE SAVING APPLIANCES (LSA)	Nominal Duration: 60 Hrs.	Unit Code: SICIP-SBD-SSF-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate lifesaving appliances (LSA). It specifically includes the task of identifying types of life saving appliances, demonstrating life jackets and immersion suits, demonstrating emergency signalling devices and explain launching procedures of lifeboats and life rafts.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify types of life saving appliances	1.1. <u>Personal life-saving appliances</u> are identified based on their purpose. 1.2. <u>General life-saving appliances</u> are identified according to their functions. 1.3. <u>Distress signaling devices</u> are recognized as essential equipment. 1.4. Launching appliances and release mechanisms for life-saving equipment are identified. 1.5. Purpose and proper use of each appliance are interpreted based onboard safety requirements.
2. Demonstrate life jackets and immersion suits	2.1 Life jacket is selected, inspected for damage, and worn properly with all straps fastened. 2.2 Whistle and light on the life jacket are identified and tested for functionality. 2.3 Immersion suit is correctly selected based on size and environmental conditions. 2.4 Immersion suit is donned within the required time, ensuring full coverage and sealed closures for water-tightness. 2.5 Mobility and functionality are demonstrated after wearing both the life jacket and immersion suit.
3. Demonstrate emergency signaling devices	3.1 Emergency signaling devices are identified and their functions are explained. 3.2 Flares are demonstrated in the correct manner according to safety guidelines. 3.3 Whistles are blown at appropriate intervals to attract attention during an emergency. 3.4 Reflective mirrors are used to reflect sunlight and signal for help. 3.5 Sound-producing devices are activated to signal distress in the event of a man-overboard situation.

	3.6 Proper handling and storage of signaling devices are demonstrated.
4. Explain launching procedures of lifeboats and life rafts	4.1 <u>Launching procedures of lifeboats and life rafts</u> are explained. 4.2 Lifeboat launching tools and equipment are identified and their operation is described. 4.3 Pre-launch checks are explained. 4.4 Launching of life rafts using hydrostatic release units or manual methods is explained. 4.5 Emergency precautions and communication procedures during launching are interpreted.

Range of Variables

Variable	Range (Includes but not limited to):
1. Personal life-saving appliances	1.1 Life Jackets (or Life Vests) 1.2 Life Buoys 1.3 Lifelines 1.4 Thermal Protective Aids 1.5 Immersion Suits 1.6 Anti exposure suit
2. General life-saving appliances	2.1 Rescue boat 2.2 Survival craft 2.2.1 Life raft 2.2.2 Life boat
3. Distress signaling devices	3.1 Hand flare 3.2 Rocket parachute flare 3.3 Smoke signal
4. Launching procedures of lifeboats and life rafts	4.1 Prepare the Lifeboat/Life Raft 4.2 Ensure Safety 4.3 Check Release Mechanism 4.4 Clear the Area 4.5 Launch the Lifeboat 4.6 Launch Life Rafts 4.7 Check for Occupants 4.8 Head to Safety

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify personal life-saving appliances based on their purpose. 1.2 Identify general life-saving appliances. 1.3 Identify launching appliances and release mechanisms for life-saving equipment. 1.4 Identify and test the whistle and light on the life jacket for functionality.
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	1.5 Identify emergency signaling devices and explain their functions for effective distress signaling. 1.6 Explain the procedures for launching lifeboats and life rafts. 1.7 Identify lifeboat launching tools and equipment, and describe their operation during emergency situations 1.8 Explain the process of launching life rafts using hydrostatic release units. 1.9 Interpret emergency precautions and communication procedures during the launching process.
2. Underpinning Skills	2.1 Performing identification of personal and general life-saving appliances based on purpose and function. 2.2 Demonstrating recognition and use of distress signaling devices. 2.3 Applying knowledge to identify launching appliances and release mechanisms for life-saving equipment. 2.4 Demonstrating correct selection, inspection, and wearing of life jackets with all straps fastened. 2.5 Performing donning of immersion suits ensuring proper size, full coverage, and water-tight seals. 2.6 Demonstrating mobility and functionality after wearing life jackets and immersion suits. 2.7 Demonstrating proper use of emergency signaling devices, including flares, whistles, mirrors, and sound-producing devices.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal life-saving appliances 4.3 Distress signaling devices 4.4 Life jackets and immersion suits 4.5 Emergency signaling devices

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified types of life saving appliances 1.2 demonstrated life jackets and immersion suits 1.3 demonstrated emergency signaling devices
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	1.4 explained launching procedures of lifeboats and life rafts
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE FIRE FIGHTING APPLIANCES (FFA)	Nominal Duration: 60 Hrs.	Unit Code: SICIP-SBD-SSF-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate firefighting appliances (FFA). It specifically includes the task of identifying common causes of shipboard fires, selecting appropriate appliances based on fire type, operating portable and fixed fire extinguishers and using hoses, nozzles and hydrants.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify common causes of shipboard fires	1.1 Electrical failures are identified and recognized as potential ignition sources. 1.2 Fuel leaks are identified and understood. 1.3 Improper storage of flammable materials is recognized. 1.4 Engine and machinery failures are identified. 1.5 Shipboard galley fire is identified and understood. 1.6 <u>Fire Triangle</u> and Fire Square are identified and explained according to standard firefighting principles. 1.7 Behavior of fire and its spread are recognized based on environmental and operational factors.
2. Select appliances based on fire type	2.1 Safety precautions are followed when selecting and using fire extinguishers. 2.2 <u>Fire classification</u> is identified and understood. 2.3 <u>Fire-fighting appliances</u> are checked for functionality, and their condition is confirmed. 2.4 <u>Color codes of fire extinguishers</u> are identified. 2.5 Appropriate fire extinguisher is selected based on the fire classification.
3. Operate portable and fixed fire extinguishers	3.1 <u>Personal protective equipment (PPE)</u> is selected and worn as per job requirement. 3.2 Portable fire extinguishers are selected based on the type of fire and fire classification. 3.3 The <u>PASS method</u> is explained and demonstrated. 3.4 Fixed fire extinguishing systems are operated following the standard operating procedure. 3.5 Fire extinguishers are maintained and inspected to ensure they are in <u>working condition</u> .
4. Use hoses, nozzles and hydrants	4.1 Safety procedures are followed. 4.2 Hydrants are selected and checked for accessibility and functionality. 4.3 Hoses are connected to the hydrants and properly secured.

	4.4 Nozzles are connected with the fire hose properly. 4.5 Fire isolating valve is adjusted as required. 4.6 Water supply is activated at the hydrant, and the hose is directed at the base of the fire. 4.7 Water stream is adjusted using the nozzle to ensure an efficient and controlled flow.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Fire Triangle	1.1 Heat 1.2 Fuel 1.3 Oxygen
2. Fire-fighting appliances	2.1 Fire extinguishers 2.2 Fire alarms 2.3 Sprinkler systems 2.4 Fire blankets 2.5 Breathing apparatus 2.6 Fuel quick closing valve 2.7 Fire hose 2.8 Fire hydrants 2.9 Fire nozzle 2.10 International shore connection 2.11 Fire doors 2.12 Fire dampers 2.13 Fire pumps 2.14 Fire monitor
3. Fire classification	3.1 Class A – Combustible materials 3.2 Class B – Flammable liquids 3.3 Class C – Flammable gases 3.4 Class D – Flammable metals 3.5 Class E – Electrical fire 3.6 Class F (K) – Cooking oils and fats
4. Color codes of fire extinguishers	4.1 Red for water 4.2 Black for CO ₂ 4.3 Blue for dry powder 4.4 Off-white for foam 4.5 Yellow for wet chemical
5. Personal protective equipment (PPE)	5.1 Apron 5.2 Helmet 5.3 Mask 5.4 Safety goggles 5.5 Ear plug 5.6 Safety hand gloves 5.7 Safety shoes
6. PASS method	6.1 Pull 6.2 Aim

	6.3 Squeeze 6.4 Sweep
7. Working condition	7.1 Serviceable 7.2 Ready for use 7.3 Operational 7.4 Fully charged 7.5 Tag/inspected

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify electrical failures and recognize them as potential ignition sources that may cause fires. 1.2 Identify fuel leaks and understand their potential to lead to fires and explosions. 1.3 Recognize improper storage of flammable materials as a fire risk. 1.4 Identify the Fire Triangle (heat, fuel, oxygen) and Fire Square (chemical reaction) 1.5 Identify and understand fire classifications (Class A, B, C, D, and K) to properly assess the type of fire. 1.6 Identify the color codes of fire extinguishers for easy recognition and selection in emergencies. 1.7 Identify hydrants based on their location and accessibility.
2. Underpinning Skills	2.1 Performing identification of common causes of shipboard fires. 2.2 Demonstrating recognition of improper storage of flammable materials and understanding fire behavior. 2.3 Applying knowledge of the Fire Triangle and Fire Square to explain fire principles and spread. 2.4 Fixing fire classification knowledge to select appropriate firefighting appliances. 2.5 Maintaining fire extinguishers by checking functionality, condition, and color codes. 2.6 Performing operation of portable and fixed fire extinguishers using the PASS method and SOPs. 2.7 Demonstrating selection and use of PPE during firefighting operations. 2.8 Applying procedures to connect hoses, nozzles, and hydrants safely. 2.9 Servicing water flow by activating hydrants and adjusting nozzle streams for controlled firefighting.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness

	3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Fire-fighting appliances

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified common causes of shipboard fires 1.2 selected appropriate appliances based on fire type 1.3 operated portable and fixed fire extinguishers 1.4 used hoses, nozzles and hydrants
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE FIRE DETECTION AND ALARM SYSTEMS	Nominal Duration: 60 Hrs.	Unit Code: SICIP-SBD-SSF-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate fire detection and alarm systems. It specifically includes the task of identifying fire detection systems, testing alarm panels and devices and resetting detection systems after testing.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify fire detection systems	1.1. <u>Types of fire detection systems</u> are identified based on their function. 1.2. Location of fire detection devices is recognized. 1.3. Maintenance requirements for fire detection systems are identified.
2. Test alarm panels and devices	2.1 Alarm panels and devices are checked for functionality. 2.2 Tests are conducted on alarm circuits to verify their operation. 2.3 Faults in alarm systems are identified and reported. 2.4 Necessary adjustments are made based on test results to ensure compliance with safety standards. 2.5 Test results are documented and stored for future reference.
3. Reset detection systems after testing	3.1 Detection systems are returned to their operational state after testing. 3.2 System parameters are restored to their original settings as per standard procedures. 3.3 A thorough check is conducted to confirm that all detection functions are active. 3.4 Reset actions are documented for verification and future reference. 3.5 Testing results are reviewed to ensure no faults are present before resetting.

Range of Variables

Variable	Range (Includes but not limited to):
1. Types of fire detection systems	1.1. Smoke detector 1.2. Heat detector 1.3. Flame detector

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify types of fire detection systems based on their function. 1.2 Recognize the location of fire detection devices within the facility or ship. 1.3 Identify the maintenance requirements for fire detection systems. 1.4 Identify faults in alarm systems during tests or routine checks. 1.5 Explain alarm systems based on test results. 1.6 Identify safety standards. 1.7 Explain system parameters to their original settings after adjustments.
2. Underpinning Skills	2.1 Performing identification of fire detection systems based on their function. 2.2 Demonstrating recognition of locations of fire detection devices. 2.3 Applying knowledge to identify maintenance requirements for fire detection systems. 2.4 Performing tests on alarm panels and devices to check functionality. 2.5 Fixing faults in alarm systems by identifying issues and making necessary adjustments. 2.6 Maintaining test results by documenting and storing them for future reference. 2.7 Performing reset of detection systems after testing to return them to operational state. 2.8 Applying procedures to restore system parameters to their original settings. 2.9 Servicing detection systems by conducting thorough checks.
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Alarm panels and devices

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified fire detection systems 1.2 tested alarm panels and devices 1.3 reset detection systems after testing
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2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio review
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: RESPOND TO FIRE EMERGENCIES	Nominal Duration: 60 Hrs.	Unit Code: SICIP-SBD-SSF-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to respond to fire emergencies. It specifically includes the task of detecting and responding early signs of fire, following emergency fire response procedures, rescuing individuals from fire-affected areas and reporting and documenting fire incidents.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Detect and respond to early signs of fire	1.1. <u>Early signs of fire</u> are promptly detected. 1.2. Immediate response actions are taken in line with standard fire safety procedures. 1.3. Emergency personnel are notified, and firefighting measures are initiated.
2. Follow emergency fire response procedures	2.1 Fire alarms and detection systems are activated to alert the crew and passengers. 2.2 Firefighting teams are mobilized and assigned specific duties based on the ship's emergency protocol. 2.3 <u>Ventilation systems</u> are shut down. 2.4 Escape routes are identified, and evacuation procedures are initiated. 2.5 Firefighting equipment is properly deployed and used by trained personnel. 2.6 <u>Communication systems</u> are maintained to coordinate between firefighting teams. 2.7 Fire situation is constantly monitored.
3. Rescue individuals from fire-affected areas	3.1 <u>Personal protective equipment (PPE)</u> is worn by rescuers. 3.2 Crew members are directed to muster stations in an orderly manner. 3.3 All crew members are accounted for at the master station. 3.4 Evacuation routes are assessed for safety and cleared. 3.5 Firefighters and emergency response teams are mobilized. 3.6 Rescue efforts are prioritized for those in immediate danger. 3.7 Injured individuals are provided with immediate first aid and assistance.
4. Report and document fire incidents	4.1 All personnel involved in the firefighting efforts are identified.

	4.2 Incident details are accurately documented.
	4.3 The ship's logbook and emergency response records are updated.
	4.4 Post-incident reports are prepared and submitted to ship management.

Range of Variables

Variable	Range (Includes but not limited to):
1. Early signs of fire	1.1. Sight view 1.2. Smoke 1.3. Heat 1.4. Unusual sound 1.5. Unusual smell
2. Ventilation systems	2.1 Fire door 2.2 Fire damper 2.3 Blower
3. Communication systems	3.1 Public Address (PA) system 3.2 Walkie talkie system 3.3 Rope pulling system
4. Personal protective equipment (PPE)	4.1 Apron 4.2 Helmet 4.3 Mask 4.4 Safety goggles 4.5 Fireman's outfit 4.6 Safety hand gloves 4.7 Safety shoes

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify early signs of fire to prevent further escalation. 1.2 Explain standard fire safety procedures. 1.3 Describe fire alarms and detection systems to alert the crew and passengers of a fire. 1.4 Explain firefighting teams and assign them specific duties according to the ship's emergency protocol. 1.5 Identify escape routes and initiate evacuation procedures. 1.6 Describe firefighters and emergency response teams to handle fire suppression and rescue operations. 1.7 Identify all personnel involved in the firefighting efforts. 1.8 Explain update the ship's logbook and emergency response records.
2. Underpinning Skills	2.1 Performing detection of early signs of fire promptly. 2.2 Applying immediate response actions according to standard fire safety procedures.

	2.3 Demonstrating notification of emergency personnel and initiation of firefighting measures. 2.4 Performing activation of fire alarms and detection systems to alert crew and passengers. 2.5 Maintaining coordination by mobilizing firefighting teams and assigning duties per emergency protocol. 2.6 Fixing ventilation systems by shutting them down and clearing escape routes for evacuation. 2.7 Performing deployment and proper use of firefighting equipment by trained personnel. 2.8 Applying first aid and assistance to injured individuals.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Ventilation systems

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 detected and responded to early signs of fire 1.2 followed emergency fire response procedures 1.3 rescued individuals from fire-affected areas 1.4 reported and documented fire incidents
2 Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio review
3 Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Ship Safety and Firefighting
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sq) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Portable Fire extinguishers (Water Type)	05 pcs
2.	Portable Fire extinguishers (Dry Powder)	05 pcs
3.	Portable Fire extinguishers (Foam Type)	05 pcs
4.	Portable Fire extinguishers (CO ₂)	05 pcs
5.	Fixed Fire extinguishers System (Sprinkler System)	01 Set
6.	Fixed Fire extinguishers System (CO ₂)	02 Set
7.	Fire hoses, nozzles, and hydrants	05 Set
8.	Fire blankets	15 pcs
9.	Life jackets	25 pcs
10.	Immersion suits	02 Set
11.	Lifeboats	01 pcs
12.	Life rafts	01 pcs
13.	Emergency signaling devices (Hand flares, Rocket Parachute, Smoke signal, whistles)	12 Set
14.	Lifeboat launching System	01 Set
15.	Life-saving rings and buoys	10 pcs
16.	Fire detection sensors (smoke, heat, flame detectors)	12 Set
17.	Alarm panels	01 Set
18.	Fire alarm system test panels	01 Set
19.	Manual call points	06 Set
20.	Fireman's outfit	02Set
21.	First Aid Box	03 Set
22.	International shore connection	01 set

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Ship Safety and Firefighting the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.